

KAVENNESH BALACHANDAR

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EDUCATION:

Master of Science in Cyber Security | Florida International University, Miami, FL.

August 2024 - April 2026

WORK EXPERIENCE:

Student Teaching Assistant | Florida International University, Miami, FL

January 2025 - Present

- Delivered hands-on cybersecurity and networking labs for **60+ students** covering **TCP/IP**, DNS, DHCP, Linux administration, network troubleshooting, packet capture, and network traffic analysis using **Wireshark**, strengthening security operations and incident investigation skills.
- Guided students in log analysis, vulnerability identification, **RBAC-based access control**, and secure system configuration using Linux CLI tools and network diagnostic utilities, improving lab completion rates by **22%** while reinforcing threat detection practices.
- Developed technical documentation, troubleshooting guides, and lab exercises aligned with ITSM workflows, **incident response procedures**, and security monitoring practices used in enterprise IT environments.

Network Security Intern | INTERN DECCAN SOLUTION, Chennai, India

December 2023 - February 2024

- Deployed and configured **Cisco Secure Firewall 3100** appliances across **3 enterprise sites**, implementing firewall policies, **access control lists (ACLs)**, network segmentation, and traffic filtering to strengthen network security and perimeter defense.
- Optimized **25+ firewall**, routing, DNS, DHCP, and VLAN configurations, improving network stability and security alert accuracy, reducing false positives by **20%**.
- Automated **Linux administration** and system monitoring tasks using **Bash** and **Python**, streamlining log collection, network monitoring, and incident response workflows, improving operational efficiency by **25%**.

PROJECTS:

Home Networking Lab - Personal Project

May 2025 - July 2025

- Built a **Debian Linux** security lab simulating enterprise environments, performing **TCP/IP configuration**, SSH access control, Linux administration, and network traffic analysis for hands-on cybersecurity testing.
- Implemented system monitoring, secure logging, and service automation using **systemd**, improving system reliability and operational visibility.
- Configured **RBAC-based access control**, security logging pipelines, and system health monitoring to support threat detection and incident investigation.

VPN Monitoring & Detection System:

January 2024 - March 2024

- Designed and implemented a **Deep Packet Inspection (DPI)** network monitoring system analyzing encrypted VPN traffic (**OpenVPN**, **WireGuard**) across **100+ sessions**, improving network visibility and threat detection.
- Integrated **ELK Stack (Elasticsearch, Logstash, Kibana)** SIEM platform for centralized log aggregation, security monitoring, and real-time alerting, reducing **MTTR from 30 minutes to 8 minutes**.
- Implemented **RBAC-based access control**, automated alert triage, and incident response workflows to improve security operations efficiency.

Enhanced Image Generator:

January 2023 - March 2023

- Developed a **Python-based** image processing pipeline using **OpenCV** and **NumPy** to preprocess and enhance **1,000+ datasets**, improving classification accuracy and data quality.
- Automated batch preprocessing pipelines and performance optimization workflows, improving data processing efficiency and scalability.

PUBLICATION:

VPN Blocker: Detection and Prevention of VPN Tunneling in Enterprise Networks.

February 2024 - March 2024

- Developed a **Deep Packet Inspection (DPI)**-based VPN detection framework identifying encrypted tunnels such as **OpenVPN** and **WireGuard**, improving network traffic visibility and threat detection accuracy (**90–95%**).
- Integrated **ELK Stack (Elasticsearch, Logstash, Kibana)** SIEM platform for centralized log aggregation, security monitoring, and incident detection, reducing unauthorized VPN usage by **50–60%**.

CERTIFICATIONS:

- Certified Network Security Practitioner (CNSP) – Merit (**October 2025**)
- Certified Cloud Security Practitioner - AWS (CCSP-AWS) – Merit (**January 2026**)

AWARDS AND EXTRA-CURRICULAR ACTIVITIES:

- Eraksha Cybersecurity Competition (Aug 2023): **1st** place nationally for solving real-world cybersecurity challenges.
- Kavach Hackathon (Jun 2023): Qualified at college level for India's leading cybersecurity innovation program.
- Amazon AppSec CTF (Sep 2025): Selected to compete globally in Amazon's Application Security Capture the Flag.

SKILLS AND STRENGTHS:

- Security Tools:** Wireshark, Nmap, Nessus, Snort, Burp Suite, Metasploit, tcpdump
- SIEM & Monitoring:** ELK Stack (Elasticsearch, Logstash, Kibana), Log Analysis, Alert Triage
- Networking:** TCP/IP, DNS, DHCP, VLAN, Routing & Switching, Subnetting, Firewall Rules, VPN (OpenVPN, WireGuard)
- Operating Systems:** Linux (Debian, Kali, Ubuntu), Windows Server, Windows 10
- Programming & Automation:** Python, Bash, SQL, Git, Docker, systemd, Incident Response, RBAC, IDS/IPS, Threat Detection